

Signal Processing in Practice

Assignment 6: Image Sharpening

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1 Introduction

The objective of this assignment is to implement image sharpening techniques using High-Boost Filtering. The core task involves enhancing the high-frequency components of an image (edges and textures) to improve visual clarity. We explore two approaches:

1. **Fixed Gain High-Boost Filtering:** Using a constant amplification factor k .
2. **Adaptive High-Boost Filtering:** Designing a spatially varying gain $k(m, n)$ that adapts based on local edge strength to prevent artifacts.

2 Methodology

2.1 High-Boost Filtering Model

The sharpening process is modeled by adding a high-pass filtered version of the image back to the original image, scaled by a gain factor k . Let $f(m, n)$ be the original input image and $h(m, n)$ be the high-pass filter kernel. The sharpened image $g(m, n)$ is given by:

$$g(m, n) = f(m, n) + k(m, n) \cdot [f(m, n) * h(m, n)] \quad (1)$$

where $*$ denotes the convolution operation. The term $[f(m, n) * h(m, n)]$ represents the high-frequency details (edges) of the image. An example is given in the Fig: 1 where different fixed values of K is used to sharpen the images.

2.2 Filter Design

We utilize the Laplacian operator as the high-pass filter $h(m, n)$ to detect edges. The kernel used is:

$$h = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix} \quad (2)$$

This filter computes the second-order derivative, yielding zero in flat areas and high magnitude responses at intensity discontinuities (edges).

2.3 Adaptive Gain Logic (Variable k)

A major limitation of a fixed gain k is that it amplifies noise in flat regions (if k is high) or causes "overshoot" and halo artifacts around strong edges. To address this, we implement a spatially varying gain $k(m, n)$.

Let $E(m, n)$ be the magnitude of the high-pass response:

$$E(m, n) = |f(m, n) * h(m, n)| \quad (3)$$

The logic for the variable gain $k(m, n)$ is designed to be inversely proportional to the edge strength $E(m, n)$:

- **Weak Edges (Low E):** We assign a **larger** k to boost subtle textures and fine details.
- **Strong Edges (High E):** We assign a **smaller** k to prevent halo artifacts and unnatural contrast.

The mapping function used in the implementation follows a decay curve (e.g., exponential or inverse rational) to transition smoothly between high gain for texture and low gain for edges. An example is given in the Fig: 2.

2.4 Post-Processing

Since the output image $g(m, n)$ may contain values outside the valid intensity range due to the addition of derivatives, we apply saturation clipping:

$$g_{final}(m, n) = \min(255, \max(0, g(m, n))) \quad (4)$$

3 Results

3.1 Visual Comparison

The following figures demonstrate the results of the sharpening algorithm.

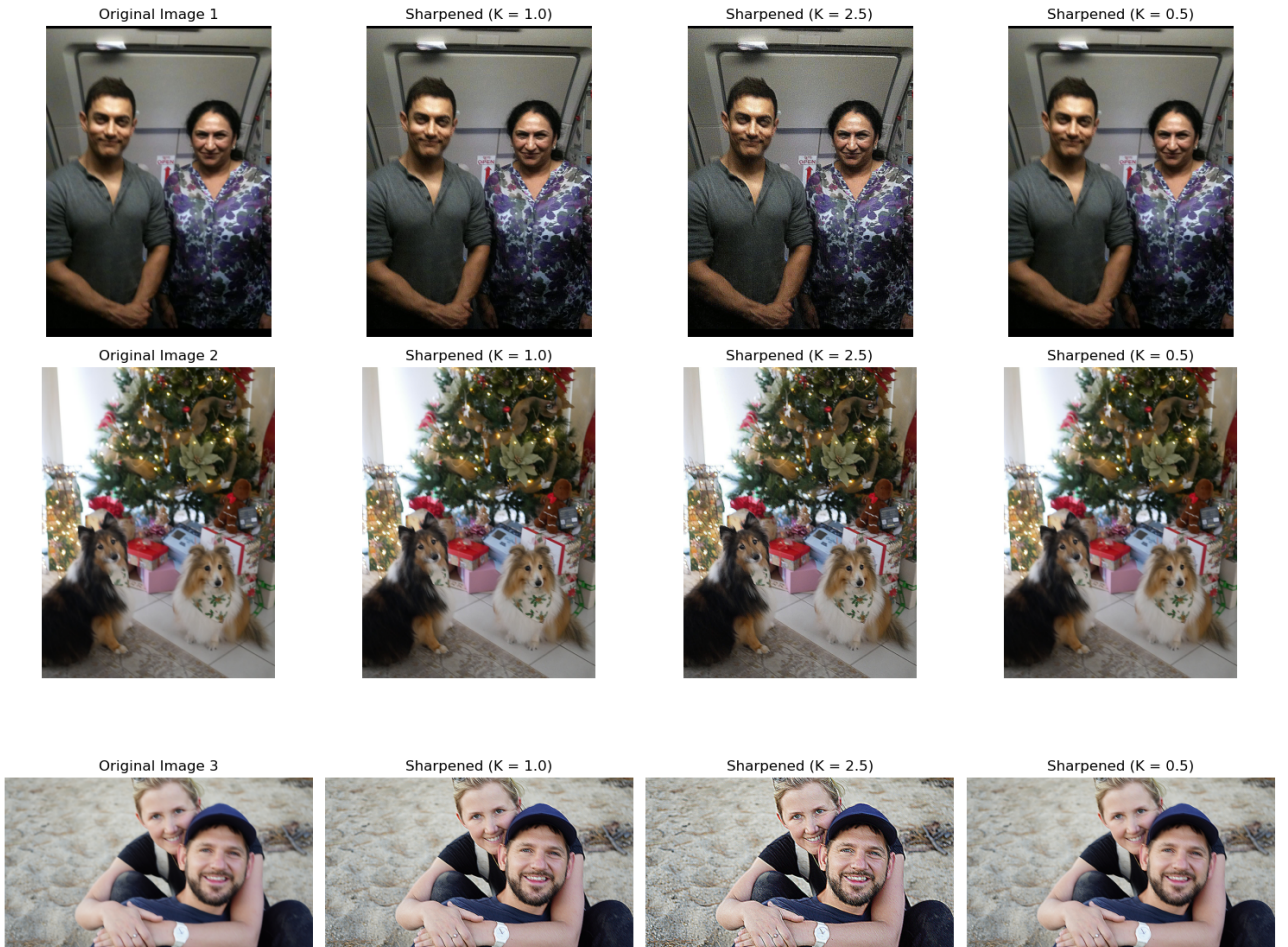


Figure 1: Comparison of Original vs Sharpened Images with different Fixed Boost(K) Values



Figure 2: Comparison of Original vs Sharpened Images with Adaptive Boost(K) Values

3.2 Gain Curve Analysis

The plot below illustrates the relationship designed between the edge strength (Laplacian magnitude) and the gain factor k .

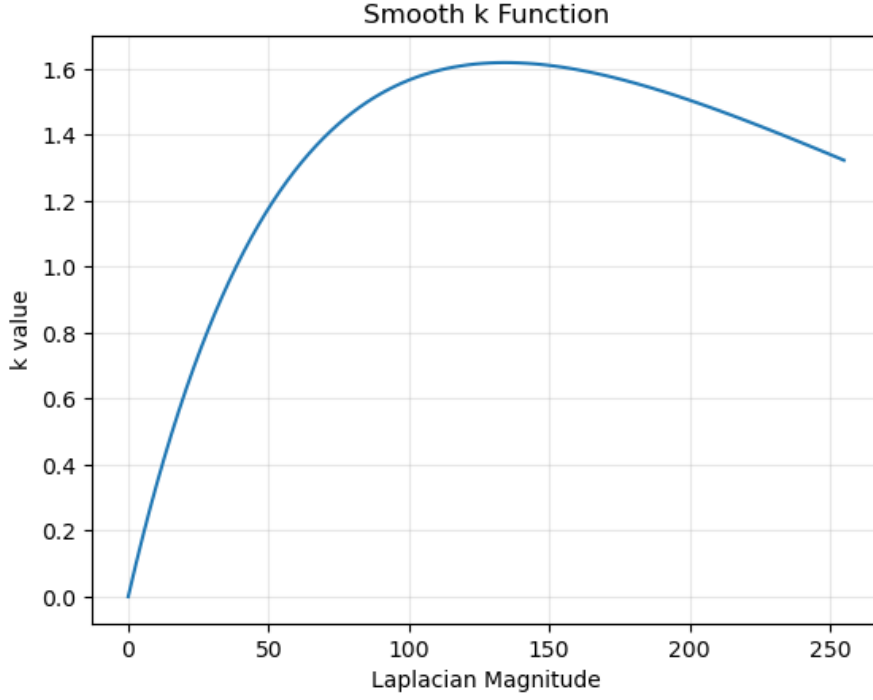


Figure 3: Adaptive Gain Curve:

The gain k decreases as the edge magnitude increases, preventing overshoot at strong edges.

4 Discussion: Variable k vs. Fixed k

4.1 Why Variable k is Superior

Based on the experimental visual results, the **Variable k (Adaptive)** approach performs significantly better than the **Fixed k** approach for the following reasons:

1. **Artifact Suppression:** Fixed high gain causes noticeable "halo" effects around strong edges (e.g., boundaries between dark and light objects). The variable k logic detects these strong edges (high Laplacian response) and automatically reduces the gain locally, preserving the natural look of the edge.
2. **Texture Enhancement:** In regions with subtle texture (low Laplacian response), a fixed low gain might be insufficient to bring out details. The variable approach allows us to set a high base gain for these weak areas without worrying about ruining the strong edges elsewhere.
3. **Noise Control:** While high-boost filtering naturally amplifies noise, the adaptive curve can be tuned to roll off quickly for very high-frequency noise or moderate edge responses, providing a better balance than a global scalar.